

Answer Key – RAG & Vector Store Fundamentals

For instructor use only

Q1 · Chunking

Q1. What is the primary reason for chunking a document before storing it in a vector store?

- (A) To reduce the file size of the original document
- ✓ (B) **To split the document into smaller pieces that fit within the LLM's context window**
- (C) To convert the document into a different file format
- (D) To remove duplicate content from the document

Correct Answer: (B)

Chunking is needed because LLMs have a limited context window. Breaking large documents into smaller pieces ensures each chunk fits within that limit and can be retrieved and passed to the LLM effectively.

Q2 · Vector Store

Q2. What does an embedding model produce when given a piece of text?

- (A) A compressed ZIP version of the text
- (B) A keyword index of the most important words
- ✓ (C) **A numerical vector that represents the meaning of the text**
- (D) A summary of the text in fewer words

Correct Answer: (C)

An embedding model transforms text into a fixed-size list of numbers (a vector) that captures its semantic meaning. Texts with similar meanings produce vectors that are close to each other in the vector space.

Q3 · RAG – Retrieval

Q3. In a RAG pipeline, what happens after the user's query is converted into an embedding?

- (A) The query is sent directly to the LLM for an answer
- ✓ (B) **The query embedding is compared against stored vectors to find the most relevant chunks**
- (C) The original documents are re-read from scratch each time
- (D) The query is stored in the vector store permanently

Correct Answer: (B)

After the query is embedded, it is compared against all stored chunk vectors using similarity search (e.g. cosine similarity). The most relevant chunks are retrieved and passed to the LLM as context.

Q4. What is the correct order of steps in a basic RAG pipeline?

- (A) Generate → Retrieve → Embed → Store
- (B) Store → Retrieve → Embed → Generate
- ✓ (C) Embed → Store → Retrieve → Generate
- (D) Retrieve → Embed → Store → Generate

Correct Answer: (C)

The correct RAG pipeline order is: (1) Embed – convert document chunks into vectors, (2) Store – save vectors in the vector store, (3) Retrieve – fetch relevant chunks for a query, (4) Generate – pass retrieved context to the LLM to produce an answer.

Q5. As part of the practical evaluation for this module, each student is required to independently develop one AI-powered application from the options listed below. Students must submit their source code.

1. Resume Analyzer
2. Chatbot
3. Email Generator
4. Content Generator

Evaluation Note:

This is a practical project submission. Evaluate based on functionality and correct use of AI concepts.